Hexuan (Matt) Wang

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Research Interests

My research lies in **Natural Language Processing and Multimodal Intelligence**, focusing on making models **reliable**: **grounded** in traceable evidence, **verifiable** by humans, and **faithful** to their own knowledge. I aim to help people use AI in research, decision-making, and creative work with understanding and trust.

Education

Johns Hopkins University Aug 2023 – May 2027 (expected)
B.S. in Computer Science and Applied Mathematics & Statistics *GPA: 3.98 / 4.00*
M.S.E. in Data Science (combined B.S./M.S.E. program)
Selected coursework: Natural Language Processing, Machine Learning, Advanced Self-Supervised Statistical Models, Multimodal Understanding & Generation, Machine Social Intelligence, Machine Programming, Matrix Analysis.

Publications & Manuscripts

[1] **Hexuan Wang**, Yaxuan Ren, Srikar Bommireddypalli, Shuxian Chen, Adarsh Prabhudesai, Rongkun Zhou, Elina Baral, Philipp Koehn. **SciTaRC: A Plan-Annotated Scientific Tabular QA Benchmark for Language Reasoning and Complex Computation.** *COLM 2026*. **[arXiv]**

[2] **Hexuan Wang**, Jingyu Zhang, Benjamin Van Durme, Daniel Khashabi. **Are Finer Citations Always Better? Rethinking Granularity for Attributed Generation.** *LCFM Workshop, NeurIPS 2026*. **[arXiv]**

[3] **Hexuan Wang**, Zixian Ma, Ranjay Krishna, Jaemin Cho. **Who Drew That Pixel? Visual-to-Code Generation and Editing through Render Attribution.** *Manuscript in preparation for CVPR 2027*.

[4] **Hexuan Wang**, Matthew Brownrigg, Christian Rispoli, Tianjian Li, Jingyu Zhang, Xi Wang, Benjamin Van Durme, Fartash Faghri, Mehrdad Farajtabar, Eric Nalisnick, Daniel Khashabi. **Chrononauts: Rethinking Temporal Evaluation with Time-Controlled Models**. *Manuscript in preparation, releasing in November*.

Research Experience

Render Attribution for Visual-to-Code Generation and Editing [3] *Mar 2026 – Present*
Advisor: Prof. Jaemin Cho | JHU CS / CLSP

- Diagnosed a **credit-assignment failure** in visual-to-code RL: GRPO gives correctly and incorrectly rendered elements the same advantage. Proposed and implemented **render attribution**, linking rendered elements to their code tokens and shared definitions to assign element-specific advantages. Across five SVG, TikZ, and HTML benchmarks, improved visual metrics over GRPO with at most 1.2% additional training-step time; code-aware credit also improved SVG editability. As an agent tool, reduced final LPIPS by **44%** and reached baseline quality with **over 95% fewer tokens**.

Chrononauts: Rethinking Temporal Evaluation with Time-Controlled Models [4] *Mar 2026 – Present*
Advisor: Prof. Daniel Khashabi | JHU CS / CLSP

- Identified a fundamental confound in historical LLM evaluation: models may recall future outcomes instead of reasoning from information available at the time. Led **Chrononauts**, a seven-cutoff (2013–2024) suite aligning model weights, post-training data, and retrieval evidence with each historical date. Engineered Wikipedia/news snapshots, cutoff-filtered SFT/RL, and leakage audits, producing 3B models competitive with larger baselines. Designed preregistered comparisons that quantify future-knowledge effects on temporal evaluations, showing that date prompts and restricted retrieval do not reliably remove them and that they grow with scale (**Llama-3.1 70B vs. 8B: +16.4 points**).

SciTaRC: Diagnosing Scientific Table Reasoning [1] *Dec 2024 – Mar 2026*
Advisor: Prof. Philipp Koehn | JHU CS / CLSP

- Conceived the **central diagnostic idea** behind SciTaRC: separating planning from execution in scientific table QA. Designed plan annotations and complexity measures to isolate these stages, and led seven collaborators in building an expert-authored benchmark. Found that oracle plans yield limited gains, revealing execution as the key bottleneck in complex scientific reasoning.

Research Experience (continued)

Citation Granularity in Attributed Generation [2] *May 2025 – Feb 2026*
Advisor: Prof. Daniel Khashabi | JHU CS / CLSP

- Designed citation-volume-controlled comparisons across 8B–120B models. Found that sentence-level citations can lower **measured attribution quality by a median 40%** with little gain in correctness, disproportionately affecting larger models and sometimes reversing benchmark rankings.

Adaptive Multimodal Grounding and Verifiable Quotations *Oct 2026 – Present*
Advisor: Prof. Jaemin Cho | JHU CS | Co-first-author manuscript planned

- Built a 70-video grounding evaluation, finding lower citation F1 on multi-video than single-video questions (Qwen3.5-9B: 15.3% vs. 30.2%). Developing **adaptive evidence acquisition** with a calibrated evidence-only verifier, so models use sufficient context while readers see **short, checkable quotations**.

Epistemic Awareness and Hallucination *2025 – 2026*
Advisor: Prof. Daniel Khashabi | JHU CS / CLSP | Pistritto Research Fellowship

- Investigated why models answer underspecified questions despite recognizing insufficient information. Identified a **representation–verbalization gap**: late-layer probes predicted unsolvability (**89.5% and 78.6% F1**), but this signal often failed to guide final answers. Tested the gap through causal interventions.

Executable User Models for Personalized Agents *2026 – Present*
Advisor: Prof. Ziyang Li | JHU CS | Co-first-author manuscript planned

- Built a neuro-symbolic agent learning user preferences from observations and corrections; led Qwen3 scaling experiments (4B–235B; **66.7% pragmatic faithfulness** vs. 59.3% raw history at 235B). Designing **executable user models** through program synthesis and verification against observed behavior.

Spectral Analysis and Reconstruction of FM Synthesis *Nov 2023 – Sep 2024*
Advisor: Prof. Daniel Naiman | JHU AMS

- Recreated DX7 FM synthesis in PyTorch; paired STFT-based audio reconstruction with interactive spectral visualizations to explain timbre. Presented at MATRX and SIAM DMV; **featured by JHU AMS**.

Baseball Scheduling Optimization *Jan 2024 – Feb 2025*
Advisor: Prof. Anton Dahbura | JHU CS / ISI

- Designed multi-objective mixed-integer scheduling models (Gurobi, Hexaly) balancing travel, fairness, rest, and franchise constraints. **Adopted in production across Minor League Baseball (MiLB).**

Honors & Awards

- **Pistritto Research Fellowship** ($6,000), JHU Computer Science (2026).
- **Naddor Prize**, JHU Applied Mathematics & Statistics; distinguished academic performance (2026).
- **Dean's List**, Johns Hopkins University (every semester).

Teaching Experience

Teaching Assistant *Jan 2024 – Present*
Johns Hopkins University

- NLP: Self-Supervised Models (F26); Artificial Intelligence (S25, F25, S26); Mathematical Foundations for CS (F24); Numerical Linear Algebra (F25); Linear Algebra & Data Science (S24, S25, F25).

Technical Skills

- **Model Training & Inference:** PyTorch, Transformers, TRL, PEFT, DeepSpeed, FSDP, vLLM, SFT, GRPO.
- **Retrieval & Evaluation:** FAISS, BM25, lm-evaluation-harness, linear probing, causal interventions.
- **Scientific Computing:** NumPy, SciPy, MATLAB, Mathematica, STFT, Gurobi, Hexaly.
- **Programming & HPC:** Python, C/C++, Bash/Linux, Slurm, Singularity, Git, LaTeX.

Academic Service & Leadership

- **Reviewer:** NeurIPS 2026.
- **JHU Vocal Chords A Cappella**, Business Manager & Bass Singer 2023 – Present
  3rd Place, ICCA Quarterfinals (2024, 2026); managed $3,000+ production and equipment budget.